CareLoop

Post-Discharge Recovery & Patient Safety Guardian

DSH HACKS V2 · AI X HEALTHCARE

Team CareLoop · Oct 2026

1. The Critical Clinical Problem

The days immediately following hospital discharge represent the highest-risk vulnerability window in the continuum of care.

- **Discharge Summary Disconnect:** Summaries are dense, abbreviated medical English. Patients and family caregivers leave confused about medication changes and red flags.
- **Medication Hazards:** Complex regimen changes frequently cause accidental double-dosing, missed critical doses, and severe OTC drug-drug interactions.
- **Silent Deterioration:** Congestive fluid overload or postoperative infection begins days before an emergency readmission, with no remote visibility.

2. The Solution: Four Continuous Care Loops

Loop 1 · DECODE (Discharge AI Parser)
Ingests unstructured discharge notes and maps to plain language in English, Bahasa Malaysia, 中文, or தமிழ் with verbatim source-span highlighting.

Loop 2 · PROTECT (Medication Guardian)
Reconciles active prescriptions against OTC items. Detects fatal drug-drug interactions (e.g. NSAID + Blood Thinner, Triple Whammy) before ingestion.

Loop 3 · WATCH (60s Check-in & Rules Engine)
Tracks weight, vitals, and symptoms. Computes an explainable 0–100 Recovery Risk Index with transparent factor attribution (Green / Amber / Red).

Loop 4 · CONNECT (Caregiver & SBAR Clinician Brief)
Automatically alerts designated family caregivers upon clinical deviation and auto-generates a one-tap clinical SBAR handover summary for doctors.

3. Verified Clinical Evidence

- **1 in 5 Medicare Patients Readmitted (19.6%):** Jencks SF, Williams MV, Coleman EA. *Rehospitalizations among patients in the Medicare fee-for-service program.* N Engl J Med. 2009;360(14):1418-1428.
- **50% Medication Discrepancies Post-Discharge:** Kripalani S et al. *Deficits in communication and information transfer between hospital and primary care physicians.* JAMA. 2007;297(8):831-841.
- **Surging Multimorbidity in SE Asia:** Institute for Public Health (IPH), National Health and Morbidity Survey (NHMS), Ministry of Health Malaysia, 2023.

Safety-First Hybrid Architecture

AI Layer: Plain-language translation & conversational comprehension.
Deterministic Rules Engine: 100% of safety decisions, risk scores, drug interactions, and emergency triggers run in auditable deterministic code. The LLM can never lower risk or alter doses.

HERO DEMO: PAK AHMAD (DAY 3 CATCH) RED ALERT (100/100)

- **Day 0-1:** Green (68.0 kg) · Routine stable
- **Day 2:** Amber (68.9 kg) · Missed furosemide dose
- **Day 3: RED (70.4 kg, +2.4 kg in 3 days)** · Orthopnoea & ankle swelling detected. System automatically triggers Caregiver Circle SMS and refreshes Clinician SBAR handover prior to emergency hospitalisation.

4. Key Technical Differentiators

- **Offline-First Demo Mode:** Runs entirely client-side with zero server dependency, zero API key requirement, and zero network data leakage.
- **Traceability & Trust:** Every plain-language item maps to exact character spans in the clinical discharge summary.
- **Live Ground-Truth Evaluation:** Evaluator computes live precision (100.0%) and recall (100.0%) — 100% F1 across 20 synthetic summaries.
- **Universal Safety Net:** Always-present emergency protocol displays immediate 999/112 dialer with patient address and active drug list.

5. Pilot Roadmap & Impact

Phase 1 (Current): Browser prototype with 68-drug database, 32 interactions, 4 languages (EN, BM, 中文, தமிழ்).
Phase 2 (Hospital Pilot): HL7/FHIR EHR connector, clinician-customizable scoring thresholds, and cellular IoT scale integration.
Projected Impact: Potential 25–35% reduction in avoidable 30-day readmissions through early outpatient intervention.